A (Square $n \times n$)

$\underbrace{A \underline{x} = \lambda \underline{x}}$ (non-zero $\underline{x}$, numbers λ).

$A x - \lambda I_n \underline{x} = \underline{0}$

$(A - \lambda I_n) \underline{x} = \underline{0}$ ✓

A has free cols iff $\det(A) = 0$.

If A has free cols; say $\underline{c}_j = \sum_{i<j} \alpha_i \underline{c}_i$

$$\det \begin{bmatrix} | & & | & | & & | \\ \underline{c}_1 & \cdots & \underline{c}_{j-1} & \underline{c}_j & \cdots & \underline{c}_n \\ | & & | & | & & | \end{bmatrix} = \det \begin{bmatrix} | & & | & | & | & & | \\ \underline{c}_1 & \cdots & \underline{c}_{j-1} & \alpha_1 \underline{c}_1 + \alpha_2 \underline{c}_2 + \cdots + \alpha_{j-1} \underline{c}_{j-1} & \underline{c}_{j+1} & \cdots & \underline{c}_n \\ | & & | & | & | & & | \end{bmatrix}$$

$$= \sum_{i<j} \alpha_i \det \begin{bmatrix} | & & | & | & | & & | \\ \underline{c}_1 & \cdots & \underline{c}_{j-1} & \underline{c}_i & \underline{c}_{j+1} & \cdots & \underline{c}_n \\ | & & | & | & | & & | \end{bmatrix} = 0.$$

If $\det(A) = 0$ then it has free cols.

OR If A has no free cols, then $\det(A) \neq 0$.

$\det(A) = \pm$ product of pivots. $\neq 0$.

depending on how many row exchanges in elimination.

<u>Check</u> $\det(A - \lambda I) = 0$.

$A = \begin{pmatrix} 1 & 0 \\ -1 & 1 \end{pmatrix}$. $\quad A - \lambda I = \begin{pmatrix} 1-\lambda & 0 \\ -1 & 1-\lambda \end{pmatrix}$.

$\lambda I = \begin{pmatrix} \lambda & 0 \\ 0 & \lambda \end{pmatrix}$.

$\det(A - \lambda I) = ?$ Use linearity properties.

First row $\underline{(1-\lambda)(1 \quad 0)} + 0(0 \quad 1)$

$$\det(A-\lambda I) = \det\begin{pmatrix} 1-\lambda\,[1 & 0] \\ -1 & (1-\lambda) \end{pmatrix} + \det\begin{pmatrix} 0\,[0 & 1] \\ -1 & 1-\lambda \end{pmatrix}.$$

$$= (1-\lambda)\underbrace{\det\begin{pmatrix} 1 & 0 \\ -1 & 1-\lambda \end{pmatrix}} + 0\,\cancel{\det}\begin{pmatrix} 0 & 1 \\ -1 & 1-\lambda \end{pmatrix}.$$

Second row: $\quad -1\,[1 \quad 0] + (1-\lambda)[0\,,\,1]$

$$(1-\lambda)\det\begin{bmatrix} 1 & 0 \\ -1\,[1\ 0] + (1-\lambda)[0\ 1] \end{bmatrix} = (1-\lambda)\det\begin{bmatrix} 1 & 0 \\ -1[1 & 0] \end{bmatrix} + (1-\lambda)\det\begin{bmatrix} 1 & 0 \\ (1-\lambda)[0. & 1] \end{bmatrix}$$

$$= (1-\lambda)(-1)\underbrace{\cancel{\det\begin{bmatrix} 1 & 0 \\ 1 & 0 \end{bmatrix}}}_{0} + (1-\lambda)^2 \underbrace{\det\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}}_{1}.$$

$$= \underbrace{(1-\lambda)^2}.$$

$$A = \begin{bmatrix} 1 & 0 \\ -1 & 1 \end{bmatrix}. \qquad \det(A-\lambda I) = \underbrace{(1-\lambda)}^2.$$

roots of $\det(A-\lambda I) = 0$ are $1, 1$.

$$A - 1I = \begin{bmatrix} 1 & 0 \\ -1 & 1 \end{bmatrix} - 1\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ -1 & 0 \end{bmatrix}.$$

$$\text{null}(A-I) = \left\{ \alpha\begin{pmatrix} 0 \\ 1 \end{pmatrix} : \alpha \in \mathbb{R} \right\}.$$

$$\text{rref}(A-I) = \text{rref}\begin{bmatrix} 0 & 0 \\ -1 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}.$$

$\underline{c2} = 0\,\underline{c1}$

$\underline{c2} - \underline{0}\,\underline{c1} = \underline{0}$

Roots of $\underbrace{\det(A-\lambda I)}_{\text{characteristic polynomial}} = 0$: Eigen values.

If a root is repeated k times in $\det(A-\lambda I) = 0$, we say that eigen value has __algebraic__ multiplicity k.

Square Symmetric real A:

$A^T = A$ eg: $A = \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}$.

$$A - \lambda I = \begin{pmatrix} 1-\lambda & -1 \\ -1 & 1-\lambda \end{pmatrix}.$$

$\det(A - \lambda I) \cancel{=} (1-\lambda)^2 - 1 = \lambda^2 - 2\lambda + 1 - 1 = \lambda^2 - 2\lambda.$

Roots: 0, 2.

$\text{null}(A - 0I) = \text{null}(A) = \left\{\alpha\begin{pmatrix}1\\1\end{pmatrix} : \alpha \in \mathbb{R}\right\}.$

$\text{rref}(A) = \begin{pmatrix} 1 & -1 \\ 0 & 0 \end{pmatrix}$. $\underline{c}2 = -\underline{c}1$

$\underline{c}2 + \underline{c}1 = \underline{0}$.

$\text{null}(A - 2I) = \text{null}\left[\begin{pmatrix} -1 & -1 \\ -1 & -1 \end{pmatrix}\right] = \left\{\alpha\begin{pmatrix}1\\-1\end{pmatrix} : \alpha \in \mathbb{R}\right\}.$

Dot product of $\begin{pmatrix}1\\1\end{pmatrix}$ and $\begin{pmatrix}1\\-1\end{pmatrix} = 0$.

The eigenvectors (one from each of null (A) and null $(A-2I)$) $\begin{pmatrix}1\\1\end{pmatrix}$ and $\begin{pmatrix}1\\-1\end{pmatrix}$ form a max linearly indpt set in $\mathbb{R}^2$.

For all $\underline{x} \in \mathbb{R}^2$, $\underline{x} = \alpha\begin{pmatrix}1\\1\end{pmatrix} + \beta\begin{pmatrix}1\\-1\end{pmatrix}$.

$A\begin{pmatrix}1\\1\end{pmatrix} = 0\begin{pmatrix}1\\1\end{pmatrix}$

$A\begin{pmatrix}1\\-1\end{pmatrix} = 2\begin{pmatrix}1\\-1\end{pmatrix}$.

$$A\underline{x} = A\left[\alpha\begin{pmatrix}1\\1\end{pmatrix} + \beta\begin{pmatrix}1\\-1\end{pmatrix}\right].$$

$$= A\left[\alpha\begin{pmatrix}1\\1\end{pmatrix}\right] + A\left[\beta\begin{pmatrix}1\\-1\end{pmatrix}\right].$$

$$= \alpha A\begin{pmatrix}1\\1\end{pmatrix} + \beta A\begin{pmatrix}1\\-1\end{pmatrix} =$$

$$= \alpha.0.\begin{pmatrix}1\\1\end{pmatrix} + 2\beta\begin{pmatrix}1\\-1\end{pmatrix}.$$

$$A(A\underline{x}) = 4\beta\begin{pmatrix}1\\-1\end{pmatrix}.$$

$$B = \begin{pmatrix} 1 \\ -1 \end{pmatrix}\begin{pmatrix} 1 & -1 \end{pmatrix} + \frac{1}{2}\begin{pmatrix} 1 \\ 1 \end{pmatrix}\begin{pmatrix} 1 & 1 \end{pmatrix} = \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix} + \begin{pmatrix} 1/2 & 1/2 \\ 1/2 & 1/2 \end{pmatrix}$$

$$= \begin{pmatrix} 3/2 & 1/2 \\ 1/2 & 3/2 \end{pmatrix}.$$

$$\lambda I = \lambda \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} \lambda & 0 \\ 0 & \lambda \end{pmatrix}$$

Eigenvalues of B.

$$\det(B - \lambda I) = \det \begin{pmatrix} 3/2-\lambda & 1/2 \\ 1/2 & 3/2-\lambda \end{pmatrix}.$$

$$\begin{bmatrix} \frac{3}{2}-\lambda & 1/2 \\ 1/2 & 3/2-\lambda \end{bmatrix} \xrightarrow[\frac{}{3/2-\lambda}]{r2 \leftarrow r2 - \frac{1}{2} r1} \begin{bmatrix} 3/2-\lambda & 1/2. \\ 0 & \left(\frac{3}{2}-\lambda\right) - \dfrac{\frac{1}{2}\cdot\frac{1}{2}}{\left(\frac{3}{2}-\lambda\right)}. \end{bmatrix} \quad \lambda \neq 3/2.$$

$$\det(B) = \left(3/2-\lambda\right)\left(\frac{3}{2}-\lambda\right) - 1/4. = \lambda^2 - 3\lambda + 9/4 - 1/4 = \lambda^2 - 3\lambda + 2.$$
$$= (\lambda-2)(\lambda-1)$$

Roots : 2, 1

$\lambda = 2$

$$B - \lambda I = \begin{pmatrix} \frac{3}{2}-\lambda & 1/2 \\ 1/2 & 3/2-\lambda \end{pmatrix} = \begin{pmatrix} -1/2 & 1/2 \\ 1/2 & -1/2 \end{pmatrix}$$

$\underline{c2} = -c1.$

$\underline{c}1 + \underline{c}2 = 0.$

$\therefore$ Eigenvector for eigenvalue $\lambda = 2$ $\underline{\begin{pmatrix} 1 \\ 1 \end{pmatrix}}.$

$$\lambda = 1 \quad B - \lambda I = \begin{pmatrix} 1/2 & 1/2 \\ 1/2 & 1/2 \end{pmatrix}$$

$\underline{c\,2} = \underline{c1}.$

Eigenvector for eigenvalue $\underline{\underline{\lambda = 1}}$ $\begin{pmatrix} 1 \\ -1 \end{pmatrix}.$

$$\underline{x} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}. = \frac{1}{2}\begin{pmatrix} 1 \\ 1 \end{pmatrix} + \frac{1}{2}\begin{pmatrix} 1 \\ -1 \end{pmatrix}.$$

$$B\underline{x} = \frac{1}{2} B \begin{pmatrix} 1 \\ 1 \end{pmatrix} + \frac{1}{2} B \begin{pmatrix} 1 \\ -1 \end{pmatrix} = \frac{1}{2}\cdot 2 \begin{pmatrix} 1 \\ 1 \end{pmatrix} + \frac{1}{2}\cdot 1 \cdot \begin{pmatrix} 1 \\ -1 \end{pmatrix}.$$

$\begin{pmatrix} 1 \\ 1 \end{pmatrix}$ is in null $(B - 2I)$ $\quad (B-2I)\begin{pmatrix} 1 \\ 1 \end{pmatrix} = 0. \quad B\begin{pmatrix} 1 \\ 1 \end{pmatrix} - 2\begin{pmatrix} 1 \\ 1 \end{pmatrix} = 0.$

$$B\begin{pmatrix} 1 \\ 1 \end{pmatrix} = 2\begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

$$B\begin{pmatrix} 1 \\ -1 \end{pmatrix} = 1.\begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

$$B(B\underline{x}) = B\left[\frac{1}{2}\cdot 2\begin{pmatrix}1\\1\end{pmatrix} + \frac{1}{2}\cdot 1\begin{pmatrix}1\\-1\end{pmatrix}\right] = \frac{1}{2}\cdot 2^2\begin{pmatrix}1\\1\end{pmatrix} + \frac{1}{2}\cdot 1^2\begin{pmatrix}1\\-1\end{pmatrix}.$$

$$B^n(\underline{x}) = \frac{1}{2}\cdot 2^n\begin{pmatrix}1\\1\end{pmatrix} + \frac{1}{2}\cdot(1)^n\begin{pmatrix}1\\-1\end{pmatrix}.$$

Spectral Decomposition Theorem.

Suppose A is square, symmetric. $n\times n$ matrix.

(i) A has n real eigenvalues (possibly repeated).

$\det(A-\lambda I)$ has n real roots (possibly repeated).

(ii) Eigenvectors corresponding to distinct eigenvalues are orthogonal.

(iii) Suppose λ^* is an eigenvalue $(A-\lambda^* I)$. has free columns.

$\dim[\text{null}(A-\lambda^* I)] = \#$ free cols in $A-\lambda^* I$

$= \#\ \lambda^*$ is repeated among roots of $\det(A-\lambda I)$

$\Rightarrow$ we get n linearly indpt eigenvectors.

$\therefore$ eigenvectors form a basis for $\mathbb{R}^n$.

we can also get n orthogonal vectors as a basis for $\mathbb{R}^n$.

Normalize all so that they have length 1.

$\underline{v}_1 \ \dots \ \underline{v}_n$.

$A\underline{v}_1 = \lambda_1\underline{v}_1$

$A\underline{v}_2 = \lambda_2\underline{v}_2$

$\vdots$

$A\underline{v}_n = \lambda_n\underline{v}_n$.

$$\underline{v}_i^T\underline{v}_j = \begin{cases}0 & \text{if } i\neq j\\ 1 & \text{if } i=j.\end{cases}$$

$$V = \begin{bmatrix} | & & | \\ \underline{v}_1 & \cdots & \underline{v}_n \\ | & & | \end{bmatrix}.$$

$$V^TV = \begin{bmatrix} ij^{\text{th}} \text{ entry} \\ = \underline{v}_i^T\underline{v}_j \end{bmatrix} = \begin{bmatrix} 1 & & 0 \\ & \ddots & \\ 0 & & 1\end{bmatrix}$$

$$V^{-1} = V^T.$$

$$AV = A\begin{bmatrix} \underline{v}_1 & \cdots & \underline{v}_n \end{bmatrix} = \begin{bmatrix} \lambda_1 \underline{v}_1 & \lambda_2 \underline{v}_2 & \cdots & \lambda_n \underline{v}_n \end{bmatrix}$$

$$= V \underbrace{\begin{bmatrix} \lambda_1 & 0 & & 0 \\ 0 & \lambda_2 & & \vdots \\ \vdots & \vdots & & \vdots \\ 0 & 0 & & \lambda_n \end{bmatrix}}_{\Lambda.}$$

$$= V\Lambda.$$

$$A \underbrace{VV^T}_{I} = V\Lambda V^T \qquad \therefore\ A = V\Lambda V^T$$